

Objective

To implement and demonstrate `ArrayIndexOutOfBoundsException` in Java using try-catch and handle an invalid array index.

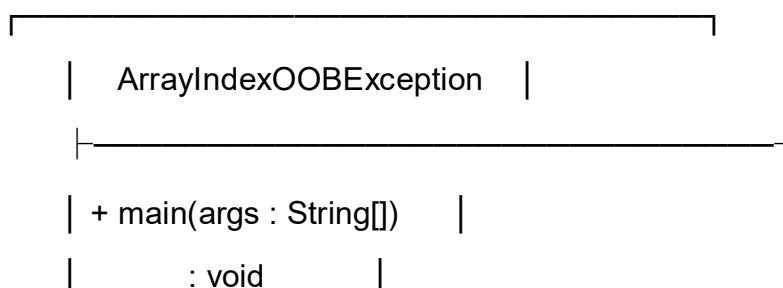
Steps for Execution

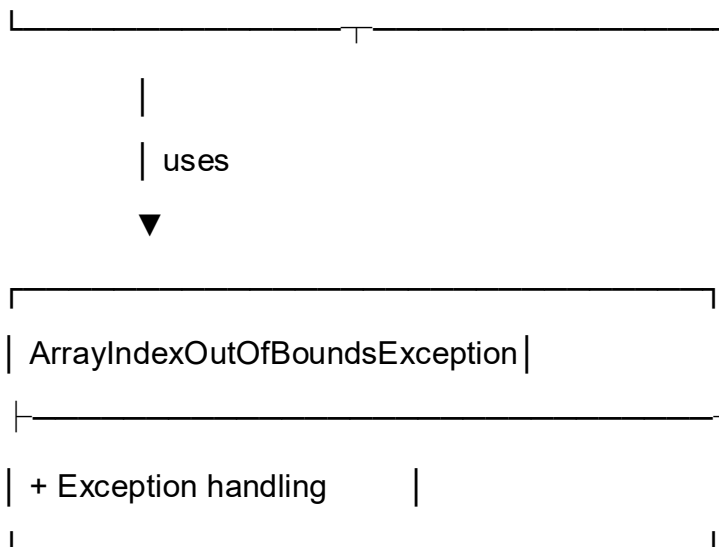
1. Create a Java project and package named `javacore`.
2. Create a file named `ArrayIndexOOBException.java` and enter the given code.
3. Create an integer array with five elements.
4. Access valid and invalid array indexes inside try.
5. Handle the exception using catch.
6. Compile and run the program.
7. Observe the output.

Algorithm

1. Start the program.
2. Create an integer array with five elements.
3. Access the first and third elements.
4. Try to access the sixth element.
5. Catch `ArrayIndexOutOfBoundsException`.
6. Display the error message.
7. Display "Program ended.".
8. Stop the program.

UML Class Diagram





Relationship

- `ArrayIndexOutOfBoundsException` contains the `main()` method.
- The program handles `ArrayIndexOutOfBoundsException` using a try-catch block.
- The exception occurs because index 5 does not exist in an array containing 5 elements.

CODE:

```
package javacore;

public class ArrayIndexOutOfBoundsException {
    public static void main(String[] args) {

        int[] marks = {80, 75, 90, 85, 70};

        try {
            System.out.println("First mark: " + marks[0]);
            System.out.println("Third mark: " + marks[2]);

            System.out.println("Sixth mark: " + marks[5]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Array index does not exist.");
        }
    }
}
```

```
    }  
  
    System.out.println("Program ended.");  
}  
}
```

Output

First mark: 80

Third mark: 90

Error: Array index does not exist.

Program ended.

Result

The program successfully demonstrates `ArrayIndexOutOfBoundsException` handling in Java. Since the array contains only five elements with indexes 0 to 4, accessing `marks[5]` causes an exception, which is handled using the catch block.